

Core Java Questions

JAVA INTERVIEW QUESTIONS

1. What is Java and why is it popular?
2. Why is Java called a platform-independent language?
3. What are the main features of Java?
4. What are data types in Java?
5. What is the difference between primitive and non-primitive data types?
6. What is a variable? Explain types of variables in Java.
7. What are conditional statements in Java?
8. What is the difference between if, if-else, and switch statements?
9. What is the difference between break and continue?
10. What is the difference between for, while, and do-while loops?
11. What is an array in Java?
12. How do you declare and initialize an array in Java?
13. What is the default value of variables in Java?
14. What is Java Architecture?
15. Explain JVM, JRE, and JDK in Java Architecture.

OOPS INTERVIEW QUESTIONS

1. What is Object-Oriented Programming (OOPS)?
2. What are the four main pillars of OOPS?
3. What is a Class and an Object?
4. What is the difference between IS-A and HAS-A relationship?
5. What is Inheritance? How does it represent IS-A relationship?
6. What is Composition in Java? How does it represent HAS-A relationship?

7. What is a Constructor in Java?
8. What are the types of Constructors in Java? What is the difference between Default Constructor and Parameterized Constructor?
9. Can constructors be inherited in Java?
10. Can we overload constructors in Java?
11. What is a Method in Java?
12. What are the types of Methods in Java? What is the difference between Static and Non-static methods?
13. What is the difference between Local Variables and Global Variables?
14. What is an Interface in Java?
15. Why do we use Interface in Java?
16. What is the difference between Interface and Abstract Class?
17. Can a class implement multiple interfaces?
18. What is the difference between Overloading and Overriding?
19. What is the difference between this and super keyword?
20. What is the difference between static block and instance block?
21. Difference between Method and Constructor?
22. Explain constructor chaining and copy constructor?

EXCEPTION HANDLING INTERVIEW QUESTIONS

1. What is an Exception in Java?
2. What is Exception Handling in Java?
3. What are the types of Exceptions in Java?
4. What is the difference between Checked and Unchecked Exceptions?
5. What is the purpose of try, catch, and finally blocks?
6. Can we use multiple catch blocks?
7. Who is the parent of all Exception classes?

8. What is the difference between throw and throws?
9. What is a Custom Exception in Java?
10. Explain the Exception hierarchy?
11. Difference between final,finally,finalize()?

THREADS & MULTITHREADING INTERVIEW QUESTIONS

1. What is a Thread in Java?
2. What is Multithreading?
3. What is the difference between Process and Thread?
4. How can you create a thread in Java?
5. What is the difference between extending Thread class and implementing Runnable interface?
6. What is the start() method and how is it different from run()?
7. What is Thread Life Cycle in Java?
8. What is Thread Priority?
9. What is Synchronization in Java?
10. What is the difference between synchronized method and synchronized block?
11. What is Deadlock in Java?
12. What is Race Condition in Java?
13. What is Inter-thread Communication?
14. What are wait(), notify(), and notifyAll() methods?
15. What is the volatile keyword in Java?
16. What is the difference between sleep() and wait() methods?

COLLECTION FRAMEWORK INTERVIEW QUESTIONS

1. What is the Collection Framework in Java?

2. What are the main interfaces in the Java Collection Framework?
3. What is the difference between Collection and Collections?
4. What is the difference between List, Set, and Map?
5. What is the difference between Array and Collection?
6. What is the difference between ArrayList and LinkedList?
7. What is the difference between Vector and ArrayList?
8. What is the difference between HashSet and TreeSet?
9. What is a Map in Java?
10. What is the difference between HashMap and Hashtable?
11. What is the difference between HashMap and TreeMap?
12. What is Iterator in Java?
13. What is the difference between Iterator and ListIterator?
14. What is the difference between Comparable and Comparator?
15. Explain hashcollision in java?

STRING INTERVIEW QUESTIONS IN JAVA

1. What is String in Java?
2. Why is String immutable in Java?
3. What is the difference between String, StringBuffer, and StringBuilder?
4. What is String Pool in Java?
5. What is the difference between == and equals() in String comparison?
6. What is the difference between equals() and compareTo() method?

7. What is substring() method in Java?
8. What is the difference between Mutable and Immutable objects?
9. How to convert String to int in Java?
10. How to split a String in Java?
11. What is classLoader in java?
12. What is the garbageCollector?